

Supplementary Material

Lineage and Release Timing in Correlated Errors of Open-Weight Language Models: A Pair-Level Study

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This supplement holds material moved from the main paper: the per-model precision analysis that motivates the pair-level design (S1), the record of findings withdrawn from the earlier version of this work (S2), the simulation check and its post-hoc audit (S3), detailed robustness results for every sample (S4), the exploratory analysis of agreement as a signal of shared lineage (S5), and the post-hoc analyses added after the third review (S6). Every table and figure is generated from the committed outputs by the repository’s table and figure scripts.

S1. Why per-model variance decomposition is not enough

Model and identification

Let Y_i be a per-model trait (e.g., accuracy), with family $f(i)$ and release quarter $e(i)$:

$$\begin{aligned} Y_i &= \mu + \alpha_{f(i)} + \beta_{e(i)} + u_i, \\ \alpha &\sim \mathcal{N}(0, \sigma_L^2), \quad \beta \sim \mathcal{N}(0, \sigma_E^2), \quad u \sim \mathcal{N}(0, \sigma_U^2), \end{aligned} \tag{1}$$

so that $\text{Var}(\mathbf{y}) = \sigma_L^2 Z_F Z_F^\top + \sigma_E^2 Z_E Z_E^\top + \sigma_U^2 I$, with only the intercept as a fixed effect. The variance components are identified when the covariance basis $\{Z_F Z_F^\top, Z_E Z_E^\top, I\}$ is linearly independent. The rank of the fixed-effects dummy design $[\mathbf{1} \mid Z_F \mid Z_E]$, used as the gate in the earlier version of this work, is not the relevant condition for this model: for the sixteen models evaluated there, that design is rank-deficient (rank 14 of 15) while the covariance basis has full rank (3 of 3). Its condition number also depends on the arbitrary choice of reference level (TABLE SI).

Precision

Identification does not imply precision. For each design we compute the expected restricted-likelihood information $I_{jk} = \frac{1}{2} \text{tr}(PV_j PV_k)$ at four share scenarios (lineage-dominant 0.5/0.2/0.3, era-dominant 0.2/0.5/0.3, balanced, and low-signal 0.1/0.1/0.8), the delta-method standard error of each share, and, as the verdict, the Monte Carlo root-mean-square error (RMSE) of the share estimates under the same REML estimator (500 replicates per scenario). The analytic standard error tracks the Monte Carlo RMSE closely (e.g., 0.289 versus 0.285 for the sixteen-model design), so it can be used before any measurement.

Table SI: Precision of family and era variance shares for per-model traits. FE: fixed-effects dummy design used as a gate in the earlier version; κ range is over every choice of reference levels. Covariance-basis rank 3/3 means the variance components are identified. The last column is the worst-case Monte Carlo RMSE of either share across four scenarios (500 replicates each).

Design	Families	Quarters	FE full rank	$\kappa(X^\top X)$ range	Cov. basis	Worst RMSE
16 evaluated models	5	11	no	singular	3/3	0.302
22-model selection	6	14	yes	285–1526	3/3	0.256
47-model candidate frame	6	14	yes	166–1116	3/3	0.189
30-model “sufficient” design	6	8	yes	91–144	3/3	0.194
42-model sweep design	7	8	yes	83–113	3/3	0.182
nested reference (72)	6	6	no	singular	2/3	not identified

No design reaches an RMSE of 0.10 for both shares (TABLE SI): not the sixteen evaluated models (0.30), not the 47-model candidate frame (0.19), and not the 30-model configuration the earlier version described as sufficient (0.19). Removing the single model responsible for the fixed-effects rank deficiency leaves the RMSE unchanged (0.302). The binding constraint is the number of *levels* of each factor: more models per level help only up to a point (with six families and fourteen quarters, 22 models give 0.256 and 47 give 0.189). In staggered designs (main paper, Fig. 4), bringing both shares within ± 0.10 requires on the order of sixty families and fourteen quarters; with eight quarters the era share never falls below about 0.14. Because release quarters are few by construction, a per-model decomposition is a poor instrument for the paper’s research questions in populations like those examined here. Within the designs and scenarios examined, this motivates the pair-level design; it does not show that the shares are inestimable under every design, only that the required numbers of families and quarters exceed those available.

S2. Changes from the earlier version

The earlier version of this manuscript (archived in the repository as `paper/src/archive/`) reported a sixteen-model evaluation and an identifiability gate. TABLE SII lists each finding of that version that is withdrawn or corrected, why, and how it is treated here.

Table SII: Findings of the earlier version and their treatment.

Earlier claim	Reason	Current treatment
Variance components “not identifiable” (fixed-effects dummy-design gate)	Random-effects model identified (basis rank 3/3); condition number depends on reference level (115–585)	Withdrawn; imprecision quantified (Section S1)
Item-level predictions; Qwen-7B accuracy 0.2294	Code read only option A’s log-likelihood; 0.2294 is the all-A rate	Withdrawn; code fixed; validation against stored correctness (main paper, Section III)
30-model design “passes” at $\kappa \leq 50$	Its κ is 93	Corrected
Sweep enforced crossing and connectedness	100 of 105 crossed, 60 connected	Corrected
LPM and GLMM both hit the era boundary in 20–60% of replicates	Only the GLMM; LPM 0–6.7%	Corrected
Nested-design detector: “zero silent coverage”	Undefined: no replicate went undetected	Corrected
Sixteen-model population as pre-specified	Four planned models not evaluated, no recorded reason; two DeepSeek models imputed	Arm not used for inference

S3. Simulation check of separability

The main paper (Section IV-F) describes the simulated data and criteria. Table SIII gives the pre-specified check (20 replicates per scenario) for three caps on the number of models per root; Table SIV gives the

post-hoc audit with 500 replicates, shown in the main paper’s Fig. 6. With 20 replicates the rates move in steps of five points, which explains why the audit ranks the caps differently. Cap 40 remains the primary design because it was the dated choice made before any pair outcome was computed.

Table SIII: Pre-specified simulation check (primary roster, 14,042 simulated items, 20 replicates per scenario). Null: largest false-rejection rate. Leakage: largest rejection of the wrong coefficient. Power: rejection of the lineage / same-month coefficient at the stated effect weight.

Cap	N	Null	Leakage	Power $\lambda=0.05$	Power $\lambda=0.03$
15	561	15%	20%	100% / 100%	50% / 75%
40	613	5%	10%	100% / 100%	40% / 95%
none	721	10%	15%	100% / 100%	95% / 95%

Table SIV: Post-hoc gate audit (500 replicates per truth; Wilson 95% intervals, in percent). Null: largest rejection rate of any term under no effect. Leakage: largest cross-term rejection across $\lambda \in \{0.03, 0.05, 0.10\}$. Power: rejection of the lineage (L) or release-month (E) term. Post hoc.

Sample	Cap	N	Null	Leakage	L, $\lambda=0.05$	E, $\lambda=0.05$	L, $\lambda=0.03$	E, $\lambda=0.03$
Primary	15	561	6.0 [4.2, 8.4]	7.2 [5.2, 9.8]	99.6 [98.6, 99.9]	100.0 [99.2, 100.0]	40.2 [36.0, 44.6]	83.8 [80.3, 86.8]
Primary	40	613	7.8 [5.8, 10.5]	10.8 [8.4, 13.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	63.2 [58.9, 67.3]	87.8 [84.6, 90.4]
Primary	none	721	6.6 [4.7, 9.1]	7.2 [5.2, 9.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	90.4 [87.5, 92.7]	90.2 [87.3, 92.5]
Expanded	40	967	5.6 [3.9, 8.0]	12.6 [10.0, 15.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	93.6 [91.1, 95.4]	97.8 [96.1, 98.8]

S4. Detailed robustness results

The main paper’s robustness table reports the primary sample. The tables below give every sample; figures show the same estimates. “Same month” is the 0-month gap bin relative to 12 or more months. S2: models with accuracy ≥ 0.30 .

Table SV: Coefficients with and without the accuracy terms (95% CIs from the delete-one-root jackknife).

Term	Primary		Expanded	
	Without	With	Without	With
Shared root	+0.283 [+0.251, +0.315]	+0.133 [+0.108, +0.158]	+0.240 [+0.205, +0.275]	+0.127 [+0.110, +0.145]
Same month	+0.087 [+0.051, +0.122]	+0.004 [−0.005, +0.013]	+0.092 [+0.065, +0.119]	+0.006 [−0.002, +0.014]
1–2 months	+0.085 [+0.048, +0.123]	0.000 [−0.007, +0.007]	+0.087 [+0.060, +0.114]	+0.002 [−0.005, +0.010]
3–5 months	+0.065 [+0.031, +0.099]	−0.001 [−0.009, +0.007]	+0.078 [+0.050, +0.107]	+0.001 [−0.007, +0.008]
6–11 months	+0.029 [+0.012, +0.046]	−0.001 [−0.008, +0.006]	+0.055 [+0.031, +0.079]	+0.002 [−0.006, +0.009]

Table SVI: Inference procedures. [†]Pre-specified. Root-level procedures treat lineage roots as the unit of dependence.

Sample	Term	Standard error	Estimate [95% CI]	<i>p</i>
Primary	same root	model-dyadic [†]	+0.133 [+0.118, +0.148]	<0.001
Primary	same root	root-dyadic	+0.133 [+0.114, +0.152]	<0.001
Primary	same root	delete-one-root jackknife	+0.133 [+0.108, +0.158]	<0.001
Primary	same month	model-dyadic [†]	+0.0039 [−0.0029, +0.0106]	0.263
Primary	same month	root-dyadic	+0.0039 [−0.0041, +0.0118]	0.343
Primary	same month	delete-one-root jackknife	+0.0039 [−0.0048, +0.0125]	0.383
Primary, S2	same root	model-dyadic [†]	+0.120 [+0.105, +0.135]	<0.001
Primary, S2	same root	root-dyadic	+0.120 [+0.103, +0.136]	<0.001
Primary, S2	same root	delete-one-root jackknife	+0.120 [+0.098, +0.142]	<0.001
Primary, S2	same month	model-dyadic [†]	−0.0145 [−0.0268, −0.0022]	0.021
Primary, S2	same month	root-dyadic	−0.0145 [−0.0269, −0.0021]	0.022
Primary, S2	same month	delete-one-root jackknife	−0.0145 [−0.0303, +0.0013]	0.072
Expanded	same root	model-dyadic [†]	+0.127 [+0.115, +0.140]	<0.001
Expanded	same root	root-dyadic	+0.127 [+0.112, +0.143]	<0.001
Expanded	same root	delete-one-root jackknife	+0.127 [+0.110, +0.145]	<0.001
Expanded	same month	model-dyadic [†]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded	same month	root-dyadic	+0.0059 [−0.0017, +0.0135]	0.128
Expanded	same month	delete-one-root jackknife	+0.0059 [−0.0023, +0.0142]	0.160
Expanded, S2	same root	model-dyadic [†]	+0.122 [+0.111, +0.133]	<0.001
Expanded, S2	same root	root-dyadic	+0.122 [+0.108, +0.136]	<0.001
Expanded, S2	same root	delete-one-root jackknife	+0.122 [+0.105, +0.139]	<0.001
Expanded, S2	same month	model-dyadic [†]	−0.0089 [−0.0185, +0.0008]	0.071
Expanded, S2	same month	root-dyadic	−0.0089 [−0.0191, +0.0013]	0.088
Expanded, S2	same month	delete-one-root jackknife	−0.0089 [−0.0218, +0.0040]	0.178

Table SVII: Outcome and weighting variants (95% CIs, model-level dyadic SEs). Every pair has at least 1,166 jointly wrong items, so the variant restricted to pairs with at least 1,000 is identical to the pre-specified fit.

Sample	Variant	Shared root	Same month	<i>p</i> (same month)
Primary	primary outcome, OLS (pre-specified)	+0.133 [+0.118, +0.148]	+0.0039 [−0.0029, +0.0106]	0.263
Primary	WLS, weight = jointly-wrong items	+0.142 [+0.125, +0.159]	+0.0117 [+0.0016, +0.0217]	0.023
Primary	pairs with ≥ 1000 jointly-wrong items	+0.133 [+0.118, +0.148]	+0.0039 [−0.0029, +0.0106]	0.263
Primary	phi of error indicators (secondary)	+0.170 [+0.149, +0.191]	+0.0184 [+0.0097, +0.0272]	<0.001
Primary	agreement minus option-distribution chance (exploratory)	+0.125 [+0.111, +0.140]	−0.0018 [−0.0046, +0.0011]	0.227
Expanded	primary outcome, OLS (pre-specified)	+0.127 [+0.115, +0.140]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded	WLS, weight = jointly-wrong items	+0.136 [+0.122, +0.151]	+0.0114 [+0.0033, +0.0195]	0.006
Expanded	pairs with ≥ 1000 jointly-wrong items	+0.127 [+0.115, +0.140]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded	phi of error indicators (secondary)	+0.163 [+0.146, +0.180]	+0.0194 [+0.0118, +0.0270]	<0.001
Expanded	agreement minus option-distribution chance (exploratory)	+0.116 [+0.105, +0.127]	+0.0007 [−0.0019, +0.0033]	0.604

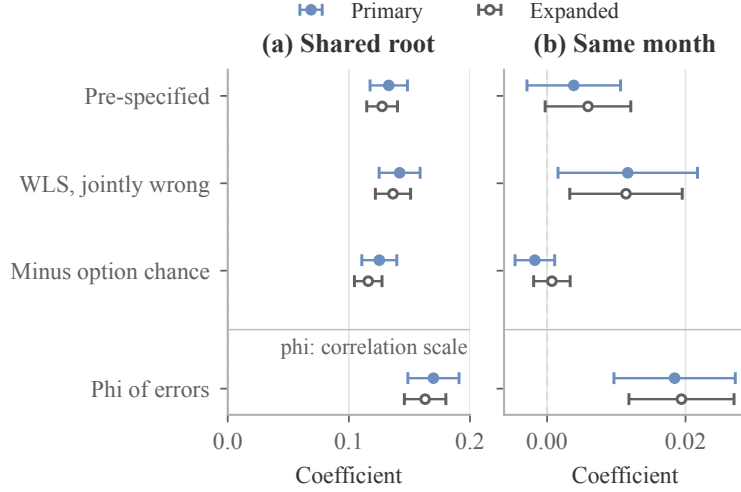


Figure S1: Outcome and weighting variants (95% CIs). Phi is a correlation, on a different scale.

Table SVIII: Item-difficulty-aware nulls (post hoc). Mean: average outcome across pairs. 95% CIs and p -values from the delete-one-root jackknife. N1a: observed agreement minus the item-level distractor null computed from all models; N1b: the same with roots weighted equally; N2: co-failure beyond a Rasch model, per item.

Sample	Outcome	Mean	Shared root	p	Same month	p
Primary	Same-wrong agreement (primary outcome)	0.423	+0.133 [+0.108, +0.158]	<0.001	+0.0039 [-0.0048, +0.0125]	0.383
Primary	N1a: minus item distractor null	0.002	+0.133 [+0.107, +0.159]	<0.001	+0.0040 [-0.0047, +0.0127]	0.367
Primary	N1b: as N1a, roots weighted equally	0.019	+0.133 [+0.108, +0.159]	<0.001	+0.0038 [-0.0049, +0.0124]	0.393
Primary	N2: co-failure over Rasch null (per item)	0.000	+0.040 [+0.032, +0.048]	<0.001	+0.0036 [+0.0012, +0.0060]	0.003
Primary, S2	Same-wrong agreement (primary outcome)	0.551	+0.120 [+0.098, +0.142]	<0.001	-0.0145 [-0.0303, +0.0013]	0.072
Primary, S2	N1a: minus item distractor null	0.000	+0.119 [+0.096, +0.142]	<0.001	-0.0138 [-0.0310, +0.0034]	0.116
Primary, S2	N1b: as N1a, roots weighted equally	0.023	+0.120 [+0.097, +0.143]	<0.001	-0.0143 [-0.0315, +0.0030]	0.104
Primary, S2	N2: co-failure over Rasch null (per item)	0.000	+0.035 [+0.028, +0.042]	<0.001	+0.0008 [-0.0019, +0.0036]	0.561
Expanded	Same-wrong agreement (primary outcome)	0.434	+0.127 [+0.110, +0.145]	<0.001	+0.0059 [-0.0023, +0.0142]	0.160
Expanded	N1a: minus item distractor null	0.004	+0.128 [+0.110, +0.146]	<0.001	+0.0061 [-0.0024, +0.0146]	0.160
Expanded	N1b: as N1a, roots weighted equally	0.024	+0.128 [+0.110, +0.146]	<0.001	+0.0062 [-0.0023, +0.0146]	0.154
Expanded	N2: co-failure over Rasch null (per item)	0.000	+0.038 [+0.032, +0.045]	<0.001	+0.0036 [+0.0012, +0.0060]	0.003
Expanded, S2	Same-wrong agreement (primary outcome)	0.547	+0.122 [+0.105, +0.139]	<0.001	-0.0089 [-0.0218, +0.0040]	0.178
Expanded, S2	N1a: minus item distractor null	0.000	+0.122 [+0.105, +0.140]	<0.001	-0.0088 [-0.0226, +0.0051]	0.214
Expanded, S2	N1b: as N1a, roots weighted equally	0.019	+0.123 [+0.106, +0.141]	<0.001	-0.0088 [-0.0229, +0.0052]	0.216
Expanded, S2	N2: co-failure over Rasch null (per item)	0.000	+0.037 [+0.031, +0.043]	<0.001	+0.0016 [-0.0028, +0.0060]	0.480

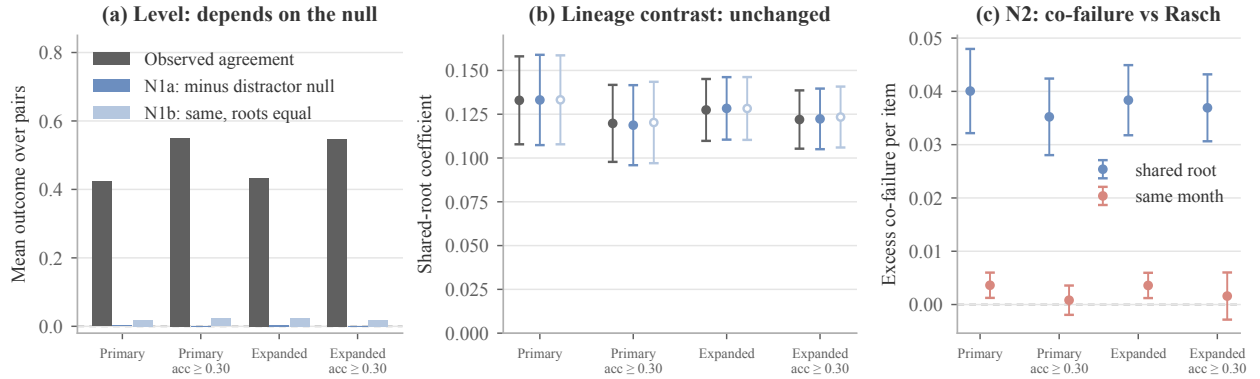


Figure S2: Item-difficulty-aware nulls. (a) Mean outcome over pairs: the level of agreement depends on the null. (b) Shared-root coefficient: the lineage contrast does not. (c) Co-failure beyond a Rasch model.

Table SIX: Size and architecture controls (exploratory; all include the S1 covariate). Entries: estimate (SE).

Sample	Models	Shared root (root-clustered SE)	Shared root (+ size, architecture)	Same month (+ size, architecture)	Same architecture
Primary	590	+0.1328 (0.0096)	+0.1256 (0.0082)	+0.0071 (0.0050)	+0.0027 (0.0026)
Primary, S2	343	+0.1166 (0.0086)	+0.1018 (0.0087)	-0.0205 (0.0110)	+0.0182 (0.0042)
Expanded	928	+0.1263 (0.0072)	+0.1201 (0.0062)	+0.0115 (0.0045)	+0.0050 (0.0022)
Expanded, S2	589	+0.1200 (0.0069)	+0.1085 (0.0057)	-0.0088 (0.0081)	+0.0181 (0.0024)

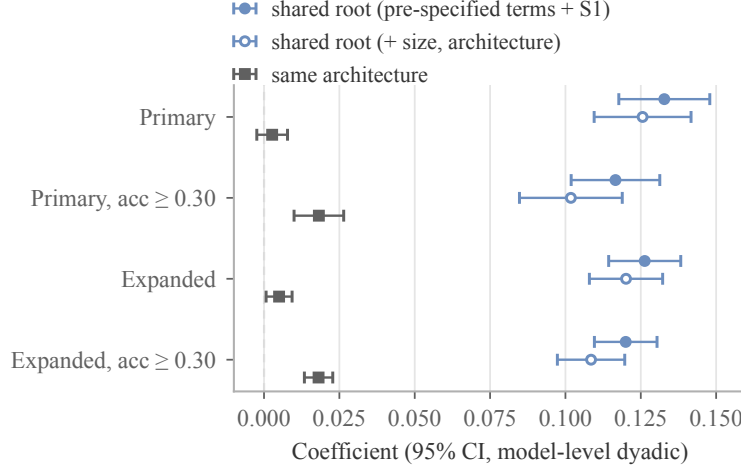


Figure S3: Size and architecture controls (95% CIs, model-level dyadic SEs).

Table SX: Further post-hoc analyses (delete-one-root jackknife 95% CIs). Equivalence bound: largest absolute same-month coefficient not excluded by the 90% interval. Root release month: gap bins from the roots' release months. CAPA: chance-adjusted agreement from chosen options. Clean sample: degenerate models removed and identical-prediction clusters collapsed. Config-consistent: models whose configuration file points to another root removed. Tree distance: primary sample only.

	Primary	Primary, S2	Expanded	Expanded, S2
Same month: 90% equivalence bound	±0.011	±0.028	±0.013	±0.020
Root release month: shared root	+0.131 [+0.108, +0.154]	+0.122 [+0.103, +0.140]	+0.119 [+0.101, +0.136]	+0.112 [+0.096, +0.127]
Root release month: same month	-0.001 [-0.011, +0.010]	-0.022 [-0.040, -0.004]	+0.009 [-0.004, +0.022]	+0.001 [-0.017, +0.019]
CAPA outcome: shared root	+0.194 [+0.153, +0.234]	+0.166 [+0.131, +0.201]	+0.183 [+0.156, +0.210]	+0.169 [+0.144, +0.195]
CAPA outcome: same month	+0.013 [+0.002, +0.023]	-0.002 [-0.013, +0.010]	+0.014 [+0.004, +0.023]	+0.002 [-0.010, +0.014]
Clean sample: shared root	+0.128 [+0.106, +0.150]	+0.117 [+0.096, +0.137]	+0.125 [+0.107, +0.142]	+0.120 [+0.104, +0.135]
Clean sample: same month	-0.002 [-0.008, +0.003]	-0.015 [-0.031, +0.001]	+0.002 [-0.005, +0.009]	-0.009 [-0.022, +0.004]
Config-consistent lineage: shared root	+0.135 [+0.108, +0.161]	-	-	-
Tree distance 1	+0.185 [+0.157, +0.213]	-	-	-
Tree distance 2	+0.136 [+0.117, +0.155]	-	-	-
Tree distance 3+	+0.115 [+0.103, +0.128]	-	-	-

S5. Agreement as a signal of shared lineage

This analysis is post hoc and descriptive. If related models share wrong answers, a pair's agreement should carry information about whether the two share a root. In the primary sample, agreement alone separates same-root from different-root pairs with an area under the receiver operating characteristic curve (AUC) of 0.95, and agreement net of the accuracy and release-gap terms of the pre-specified model with 0.88 (Fig. S4). Above an agreement of 0.7 lie 68% of same-root pairs but only 2.1% of different-root pairs. Because different-root pairs outnumber same-root pairs by 85 to 1, however, most pairs above that threshold are still unrelated (about 3,600 against 1,400), so agreement on one benchmark flags candidate relatives rather than establishing them. AUC here is a descriptive summary, not the accuracy of a validated lineage detector.

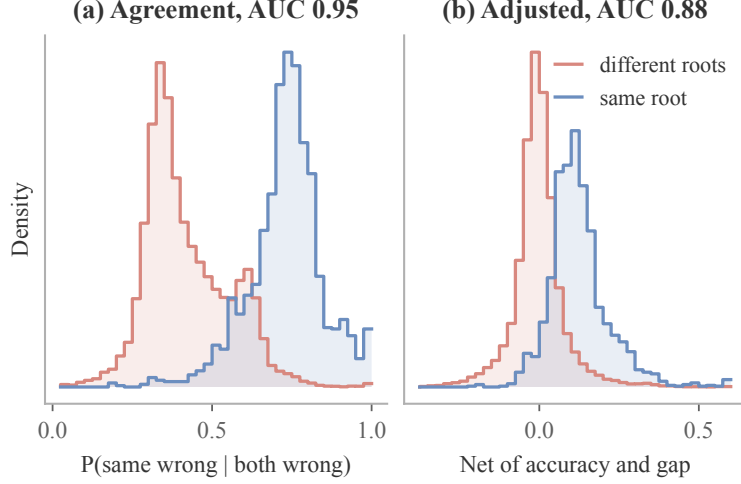


Figure S4: Agreement as a signal of shared lineage (primary sample; 2,020 same-root and 171,735 different-root pairs; densities). (a) Observed agreement on wrong answers. (b) Agreement after removing the accuracy and release-gap terms of the pre-specified model. AUC: probability that a random same-root pair scores higher than a random different-root pair.

S6. Post-hoc analyses from the third review

All analyses in this section were run after the main results were known and were not specified in advance. Code: `scripts/run_exp04_revision3.py`; outputs: `results/exp04_revision3/`.

Flexible accuracy control and answer-key errors

The pre-specified model controls accuracy through the linear sum and absolute difference of the two models' accuracies. TABLE SXI replaces them with a full quadratic surface, with fixed effects for every unordered pair of accuracy bins (deciles: 55 cells; 20 bins: 210 cells), and with the subset of pairs whose accuracies differ by at most 0.02 (plus decile cells). The answer-key check matches 5,698 of the 14,042 items to MMLU-Redux 2.0, which re-annotates 100 items per subject; 369 of the matched items are annotated as erroneous (wrong ground truth 106, no correct answer 36, several correct answers 39, unclear question or options 156, expert knowledge needed 32) and 5,329 as correct. Accuracy is recomputed on each item subset.

Table SXI: Accuracy specification and answer-key errors (post hoc; delete-one-root jackknife 95% CIs).

	Primary	Primary, S2	Expanded	Expanded, S2
<i>Shared root</i>				
Pre-specified (linear accuracy terms)	+0.133 [+0.108, +0.158]	+0.120 [+0.098, +0.142]	+0.127 [+0.110, +0.145]	+0.122 [+0.105, +0.139]
No accuracy terms (total association)	+0.283 [+0.251, +0.315]	+0.185 [+0.152, +0.218]	+0.240 [+0.205, +0.275]	+0.184 [+0.163, +0.206]
Quadratic accuracy surface	+0.123 [+0.094, +0.153]	+0.115 [+0.088, +0.141]	+0.117 [+0.101, +0.134]	+0.118 [+0.100, +0.136]
Accuracy cells: deciles (55 cells)	+0.127 [+0.096, +0.158]	+0.113 [+0.078, +0.148]	+0.125 [+0.104, +0.145]	+0.124 [+0.104, +0.143]
Accuracy cells: 20 bins (210 cells)	+0.119 [+0.084, +0.153]	+0.110 [+0.079, +0.141]	+0.118 [+0.099, +0.137]	+0.117 [+0.098, +0.136]
Accuracy-matched pairs ($ \Delta \leq 0.02$) + decile cells	+0.148 [+0.107, +0.189]	+0.134 [+0.098, +0.169]	+0.149 [+0.116, +0.181]	+0.133 [+0.102, +0.163]
MMLU-Redux: flagged items removed	+0.134 [+0.109, +0.159]	+0.121 [+0.098, +0.143]	+0.128 [+0.110, +0.146]	+0.122 [+0.106, +0.139]
MMLU-Redux items only	+0.131 [+0.105, +0.156]	+0.118 [+0.095, +0.140]	+0.128 [+0.109, +0.147]	+0.122 [+0.105, +0.139]
MMLU-Redux items annotated correct only	+0.133 [+0.107, +0.159]	+0.119 [+0.096, +0.143]	+0.130 [+0.111, +0.149]	+0.124 [+0.107, +0.140]
<i>Same month</i>				
Pre-specified (linear accuracy terms)	+0.004 [-0.005, +0.013]	-0.015 [-0.030, +0.001]	+0.006 [-0.002, +0.014]	-0.009 [-0.022, +0.004]
No accuracy terms (total association)	+0.087 [+0.051, +0.122]	+0.102 [+0.054, +0.150]	+0.092 [+0.065, +0.119]	+0.086 [+0.034, +0.138]
Quadratic accuracy surface	+0.004 [-0.004, +0.013]	-0.012 [-0.028, +0.003]	+0.006 [-0.002, +0.014]	-0.010 [-0.023, +0.004]
Accuracy cells: deciles (55 cells)	0.000 [-0.006, +0.006]	+0.002 [-0.010, +0.015]	+0.001 [-0.005, +0.008]	+0.003 [-0.007, +0.012]
Accuracy cells: 20 bins (210 cells)	-0.004 [-0.010, +0.002]	-0.012 [-0.026, +0.001]	-0.001 [-0.007, +0.005]	-0.008 [-0.020, +0.003]
Accuracy-matched pairs ($ \Delta \leq 0.02$) + decile cells	+0.033 [+0.014, +0.052]	+0.034 [-0.006, +0.074]	+0.036 [+0.011, +0.060]	+0.056 [+0.023, +0.089]
MMLU-Redux: flagged items removed	+0.004 [-0.005, +0.013]	-0.015 [-0.031, +0.001]	+0.006 [-0.002, +0.014]	-0.009 [-0.022, +0.004]
MMLU-Redux items only	+0.004 [-0.004, +0.013]	-0.014 [-0.031, +0.003]	+0.007 [-0.002, +0.015]	-0.010 [-0.023, +0.004]
MMLU-Redux items annotated correct only	+0.005 [-0.004, +0.013]	-0.015 [-0.032, +0.002]	+0.007 [-0.002, +0.015]	-0.010 [-0.024, +0.004]

Item fixed effects, bootstrap and permutation

The pair-level outcome conditions on items both models get wrong, so a pair of strong models is compared on harder items than a pair of weak models. The pair $\times$ item model regresses the indicator of the same wrong option on the pair covariates over every (pair, item) cell in which both models are wrong, with a fixed effect for each item; it is computed exactly from item-level aggregates and weights each pair by its number of jointly wrong items, so it is shown beside a weighted pair-level fit without item effects. The root cluster bootstrap resamples lineage roots with replacement (999 draws). The permutation test shuffles root labels among the models uploaded in the same quarter, preserving root sizes and the time structure (999 permutations, so the smallest attainable p is 0.001). Across the 999 permutations the largest absolute shared-root coefficient is 1.2 points in the primary sample and 1.4 in the expanded sample. The bootstrap interval is asymmetric because resampling roots sometimes omits the two largest roots, whose coefficients are among the smallest.

Table SXII: Item fixed effects and resampling inference (post hoc; 95% CIs).

	CI	Primary		Expanded	
		Shared root	Same month	Shared root	Same month
Item fixed effects	jackknife	+0.142 [+0.107, +0.176]	+0.011 [−0.001, +0.024]	+0.136 [+0.116, +0.157]	+0.011 [+0.001, +0.022]
Same weights, no item effects	jackknife	+0.142 [+0.108, +0.176]	+0.012 [−0.001, +0.024]	+0.136 [+0.116, +0.157]	+0.011 [+0.001, +0.022]
Root cluster bootstrap	bootstrap	+0.133 [+0.114, +0.196]	+0.004 [−0.003, +0.014]	+0.127 [+0.106, +0.162]	+0.006 [−0.002, +0.015]
Root-label permutation	permutation	$p = 0.001$	—	$p = 0.001$	—

Dose-response within roots, and every root counted once

Parent–child pairs come disproportionately from small roots (82 of 177), and small roots show larger coefficients, so the tree-distance gradient could reflect root composition. The within-root model uses same-root pairs only, with a fixed effect for each root, so each distance class is compared with distance 3+ pairs of the same root. The pre-specified coefficient weights roots by their number of same-root pairs, so the two largest roots dominate it; the last two rows of TABLE SXIII count roots instead.

Table SXIII: Within-root dose-response and per-root pooling (primary sample; post hoc). Within-root rows: delete-one-root jackknife 95% CIs over the roots contributing same-root pairs. Random-effects pooling uses DerSimonian–Laird weights with model-level dyadic SEs and is restricted to roots with at least 10 same-root pairs, because a root with a single same-root pair fits it exactly and has no usable SE. Unweighted mean: SE from the spread across roots.

	Estimate [95% CI]	
Within root: tree distance 1 (vs 3+)	+0.055 [+0.041, +0.068]	51 roots
Within root: tree distance 2 (vs 3+)	+0.023 [+0.003, +0.042]	51 roots
Within root: ancestor–descendant (vs more distant)	+0.047 [+0.030, +0.063]	51 roots
Within root: siblings (vs more distant)	+0.022 [+0.002, +0.041]	51 roots
Per-root coefficients, random-effects pooled (≥ 10 same-root pairs)	+0.166 [+0.125, +0.208]	16 roots; $\tau = 0.079$, $I^2 = 0.94$
Per-root coefficients, unweighted mean	+0.253 [+0.211, +0.295]	63 roots

Simulation check with the real accuracies

The original check drew abilities independently of root and month. Here each model’s ability is set so that its expected simulated accuracy equals its observed accuracy, so ability clusters by root and month exactly as in the data; the roster is the analysed primary sample (590 models), with 1,500 items and 200 replicates per scenario. A further scenario has no lineage or release-time effect, but the probability of choosing the common distractor rises with ability ($0.05 + 0.5 \text{ acc}^2$), so agreement depends non-linearly on accuracy. Each replicate is fitted with the pre-specified model and with decile accuracy cells.

Table SXIV: Rejection rates (%) with real accuracies (Wilson 95% intervals; model-level dyadic SEs; nominal 5%).

Scenario	Pre-specified accuracy terms		Decile accuracy cells	
	Shared root	Same month	Shared root	Same month
No effect	5.0 [2.7, 9.0]	8.5 [5.4, 13.2]	5.0 [2.7, 9.0]	7.0 [4.2, 11.4]
Release-time $\lambda_E = 0.03$	6.5 [3.8, 10.8]	28.0 [22.2, 34.6]	5.5 [3.1, 9.6]	27.5 [21.8, 34.1]
Lineage $\lambda_L = 0.03$	14.5 [10.3, 20.0]	7.5 [4.6, 12.0]	15.0 [10.7, 20.6]	7.0 [4.2, 11.4]
Release-time $\lambda_E = 0.05$	5.0 [2.7, 9.0]	96.0 [92.3, 98.0]	4.5 [2.4, 8.3]	97.5 [94.3, 98.9]
Lineage $\lambda_L = 0.05$	52.5 [45.6, 59.3]	6.0 [3.5, 10.2]	52.0 [45.1, 58.8]	7.0 [4.2, 11.4]
Release-time $\lambda_E = 0.10$	6.0 [3.5, 10.2]	100.0 [98.1, 100.0]	7.5 [4.6, 12.0]	100.0 [98.1, 100.0]
Lineage $\lambda_L = 0.10$	100.0 [98.1, 100.0]	6.0 [3.5, 10.2]	100.0 [98.1, 100.0]	5.0 [2.7, 9.0]
Ability-dependent distractor, no effect	27.0 [21.3, 33.5]	7.0 [4.2, 11.4]	9.0 [5.8, 13.8]	16.0 [11.6, 21.7]